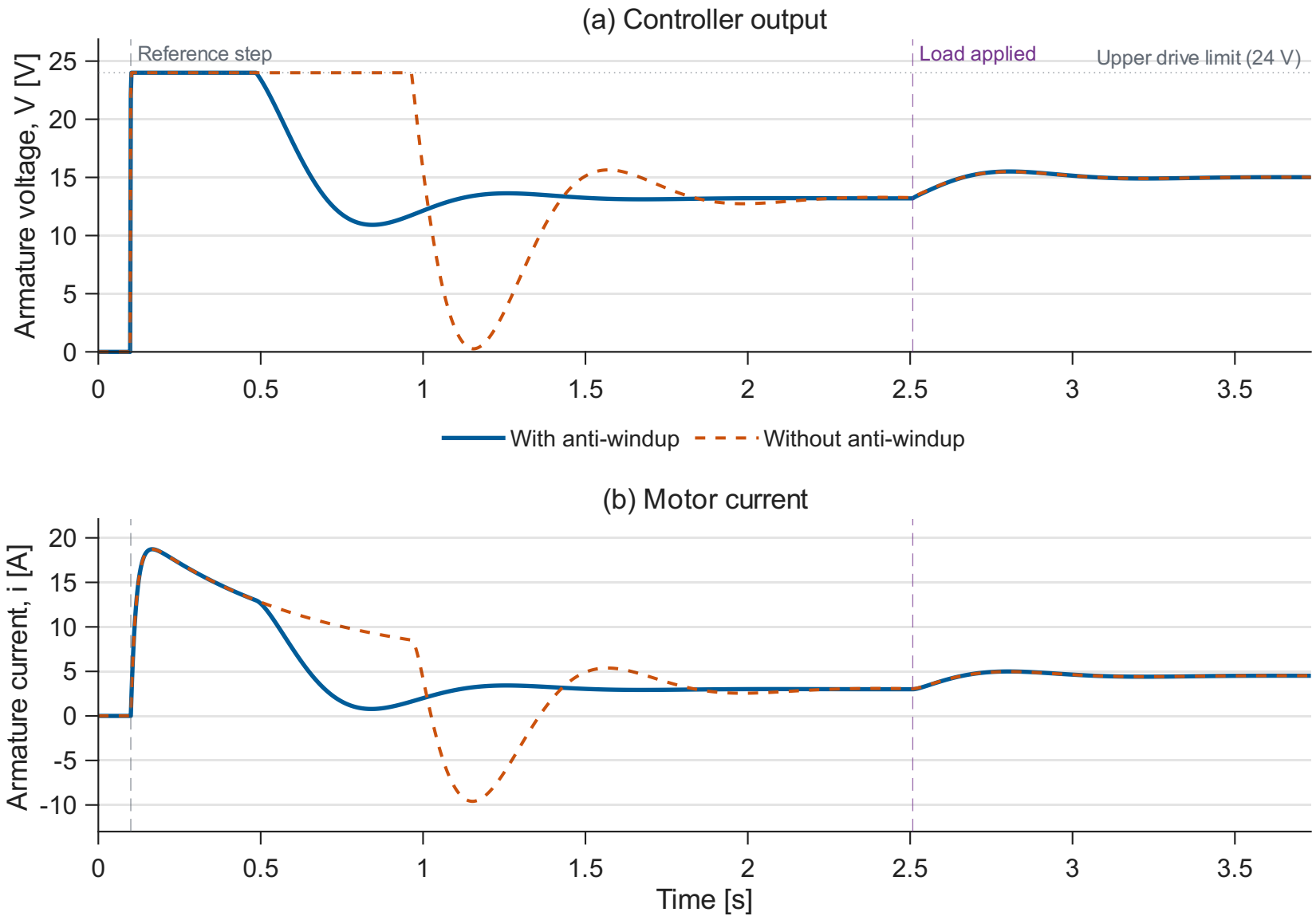

(a) Controller output
Reference step
Load applied
Upper drive limit (24 V)
Armature voltage, V [V]
25
20
15
10
5
0
0
0.5
1
1.5
2
2.5
3
3.5
With anti-windup
Without anti-windup
(b) Motor current
Armature current, i [A]
20
15
10
5
0
-5
-10
0
0.5
1
1.5
2
2.5
3
3.5
Time [s]